Birat Gautam

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PROFILE

Pursuing BSc (Hons) in Computer Science with AI. I began my programming journey with Python and have explored Machine Learning in depth, building several projects over the years. I am curious about understanding how things work under the hood. Determined to become an AI/ML Engineer. Passionate about Full Stack and Data Engineering.

SKILLS

- Python, Typescript, Next.js, Django, Fast API, Airflow, MLFlow, LangChain, Snowflake, Tensorflow, Pytorch, Numpy, Pandas, OpenCv CNN, SQL, Weaviate, Docker, Git

PROJECTS

- **Attendance Management System Using Computer Vision| React, Django, Web Sockets, OpenCv, Tensorflow, NumPy**

  Led a team of five developers to create a web application that automated attendance taking using face recognition. [URL: Click Here]

- **MatchVision – End-to-End EPL Prediction Platform | Airflow, Redis, PostgreSQL, Redis, Docker, DRF, MLFlow, Node.js, React.js**

  Containerized ML pipeline using *Airflow*, *MariaDB*, *Redis*, and *DVC* to automate data ingestion, cleaning, validation, and model training for EPL match prediction. Tracked experiments with *MLflow* and served real-time predictions via *DRF*, integrated into a *Django* web app displaying live stats, fixtures, and betting probabilities. [URL: Click Here]

- **Baali Bigyan: AI-Powered Agriculture App| Pytorch, OpenCv, YOLO, ReactNative, Django, PostgreSQL**

  Developed a mobile application leveraging *CNN* (*ResNet32*) to detect diseases in paddy plants from images, provide treatment recommendations, video tutorials, and in-app product purchases. Integrated an expert consultation feature to connect farmers with agro-vet professionals, enhancing accuracy and trust in disease diagnosis and crop management. [URL: Click Here]

- **Automated Nuclei Segmentation in Microscopy Images | CNN, TensorFlow, UNet Architecture, OpenCv**

  A *U-Net*-based deep learning model for nuclei detection in microscopy images using the 2018 Data Science Bowl dataset, achieving an IoU score of 0.88. Automated segmentation reduces manual annotation, handles complex cellular structures, and enhances diagnostic efficiency, with potential applications in resource-constrained medical and research environments. [URL: Click Here]

- **AI Powered Document RAG System | FastAPI, Weaviate, LangChain ,Redis, Next.js, PostgreSQL**

  Developed a modular backend application that enables intelligent document-based conversations using a custom Retrieval-Augmented Generation (RAG) pipeline. Document Ingestion API to upload and process PDF/TXT files, extract text, apply multiple chunking strategies, and generate embeddings stored in Weaviate with metadata in PostgreSQL. Designed a Conversational RAG API with custom retrieval and response logic (no RetrievalQAChain), integrated Redis for multi-turn chat memory. [URL: Click Here]

EDUCATION

BSc (Hons) Computer Science with Artificial Intelligence **2023 - 2027**
Birmingham City University – Birmingham, United Kingdom

High School | Science 3.78 GPA **2020 - 2022**
Uniglobe SS/College Kamaladi

AWARDS / CERTIFICATIONS AND ACTIVITIES

ICT Award 2024

Top 5 Finalists in the Rising Start Innovation Category

- Our project *Baali Bigyan* was selected as the finalist in the *ICT Award Rising Star Innovation 2024*

Sunway College Innovation Fest 2024 **July 2024**

Sunway College Kathmandu – Maitidevi, Kathmandu

- Among the 10 teams, our team was awarded the Gold Medal for our Project Attendance Management System Using Face Recognition. I led a team of 5 and contributed as a Backend Developer.

Diploma in Python Programming **July 2023**

Deerwalk Training Center – Sifal Kathmandu

- During the 4-month Diploma Course I learned the intermediate Python, Django framework, and GitHub.

UGSS Achiever Award **April 2022**

Uniglobe SS/College – Kamladi, Kathmandu (GPA: 3.78)

- Awarded Best Student of the Year for the 2020-2022 batch for my excellent performance in both academic and non-academic tasks.

CERTIFICATIONS AND ACTIVITIES

OpenEDG Python Institute: Programming with Python Professional Certificate

OpenEDG Python Institute [*Credential_Link*]

JavaScript Foundations Professional Certificate by Mozilla

Mozilla [Credential_Link]

Docker Foundations Professional Certificate

Docker, Inc [*Credential_Link*]

PUBLICATIONS

International Journal on Engineering Technology (InJET) ***May 2025***

Automated Nuclei Detection in Microscopy Images

- *Published in InJET. Developed a U-Net-based deep learning model for nuclei detection in microscopy images (2018 Data Science Bowl dataset), achieving IoU 0.88. The approach automates segmentation, improves accuracy over manual annotation, and enhances diagnostic capabilities in resource-constrained medical environments.*
- *DOI: https://doi.org/10.3126/injet.v2i2.78595*